

Simplifying expressions involving multiplication

To simplify an expression that involves terms linked by multiplication signs, the individual numbers and symbols first need to be separated from each other.

$$6a \times 2b$$

The term **6a** means $6 \times a$.
Similarly, the term **2b** means $2 \times b$.

$$6 \times a \times 2 \times b$$

Separate the expression into the individual numbers and symbols involved.

$$12 \times ab = 12ab$$

The product of multiplying 6 and 2 is 12, and that of multiplying a and b is ab . The simplified expression is $12ab$.

simplified expression written without multiplication signs

Simplifying expressions involving division

To simplify an expression involving division, look for any possible cancellation. This means looking to divide all terms of the expression by the same number or letter.

$$6pq^2 \div 2q \rightarrow \frac{6 \times p \times q \times q}{2 \times q} \rightarrow \frac{\cancel{6} \times p \times \cancel{q} \times q}{1 \times \cancel{2} \times \cancel{q} \times 1} \rightarrow \frac{3pq}{1} = 3pq$$

Annotations: q^2 means $q \times q$; this means $2 \times q$; same as $\div$; divide 6 by 2, leaving 3; divide q by q , leaving 1; divide 2 by 2, leaving 1; divide q by q , leaving 1; $3pq$ divided by 1 is simply $3pq$.

Any chance to cancel the expression down makes it smaller and easier to understand.

Both terms (top and bottom) are canceled down by dividing them by 2 and q .

Canceling down by dividing each term equally makes the expression smaller.

Substitution

Once the value of each symbol in an expression is known, for example that $y = 2$, the overall value of the expression can be found. This is called "substituting" the values in the expression or "evaluating" the expression.

Substitute the values in the expression $2x - 2y - 4y + 3x$ if

$$x = 1 \text{ and } y = 2$$



Substituting values

The formula for the area of a rectangle is $\text{length} \times \text{width}$. Substituting 5 in for the length and 8 in for the width, gives an area of $5 \text{ in} \times 8 \text{ in} = 40 \text{ in}^2$.

$$2x - 2y - 4y + 3x \rightarrow 5x - 6y \rightarrow \begin{aligned} 5x &= 5 \times 1 = 5 \\ -6y &= -6 \times 2 = -12 \end{aligned} \rightarrow 5 - 12 = -7$$

Annotations: like terms; grouped terms are easier to substitute; substitute 1 for x ; substitute 2 for y ; answer is -7 .

Group like terms together to simplify the expression.

The expression has now been simplified.

Now substitute the given values for x and y .

The final answer is shown to be -7 .